Evaluating imputation strategies for missing data through statistical modeling and simulations

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Abstract

The presence of missing data in databases is a common issue that can hinder statistical analyses, which are typically performed on complete datasets. One potential solution is data imputation, which involves estimating and filling in missing values. This study aims to evaluate the performance of imputation techniques that utilize statistical regression and to propose a practical guide for researchers based on both literature and simulation results. Our methodology involved creating artificial databases to simulate various missing data scenarios, allowing for a comprehensive analysis of different imputation techniques. We examined the impact of the proportion of missing data, the nature of the incomplete variable, and the number of incomplete variables. By adjusting linear regression models, we compared single and multiple imputation methods, along with techniques based on measures of central tendency and predictive modeling. Additionally, we assessed the impact on analysis when observations with missing data were either ignored or deleted, contrasting these outcomes with those obtained from applying imputation methods. The findings indicate that regression models are effective for data imputation, highlighting the importance of selecting appropriate methods based on the nature and extent of missing data.

Keywords: Missing data; Imputation methods; Single imputation; Multiple imputation; Statistical regression; Predictive modeling; Simulation.

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